

EXPIREMENT-1

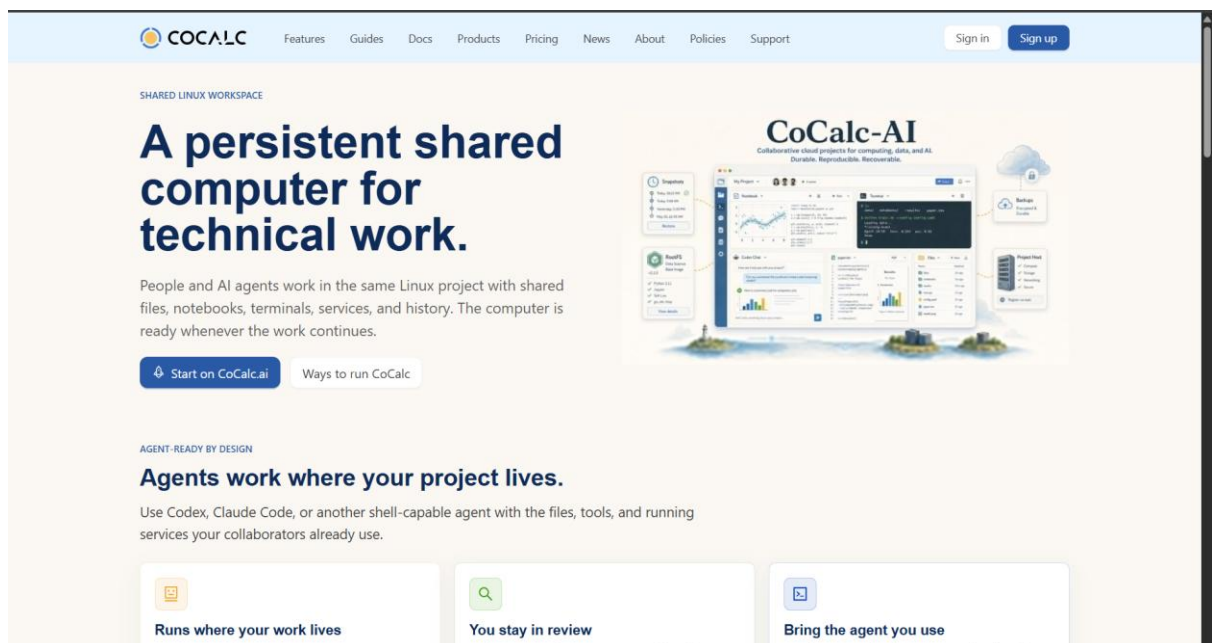
Linux Programming Environment & Basic Commands

A) Setting up of Linux Programming Environment

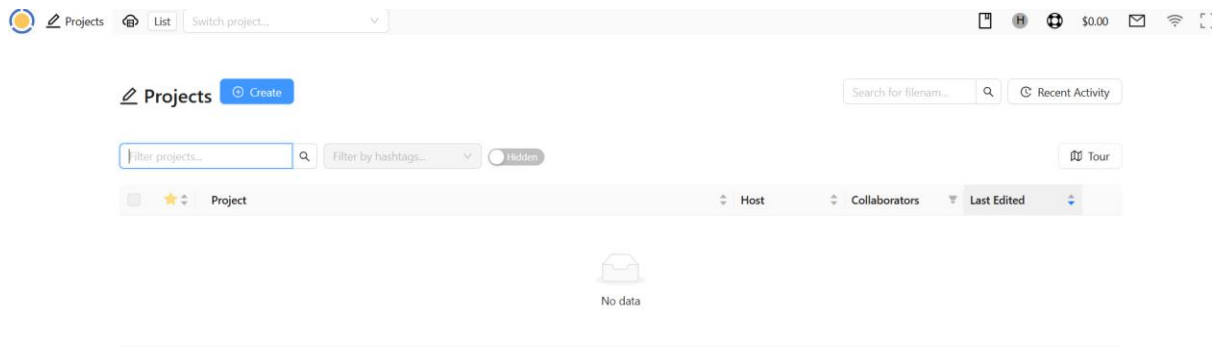
Online:

Cocalc.ai Steps:

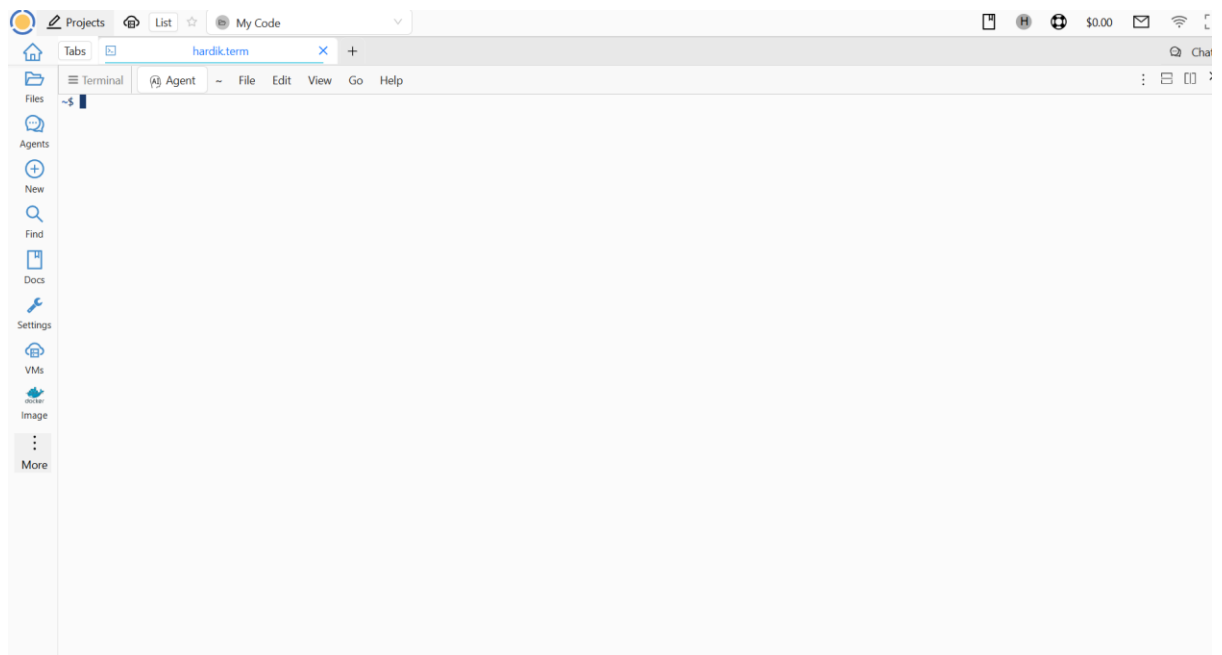
1. Open a web browser and navigate to cocalc.ai.



2. Log in or create an account.



3. Open a new C programming workspace and create a file ending in .c (e.g., main.c).



4. Write and execute C programs using the integrated web terminal (`gcc main.c -o main && ./main`).

Offline Methods:

- **Dual Boot:** Installing Linux alongside Windows on separate disk partitions, selecting the operating system at startup.
- **Single Boot:** Replacing the main operating system completely with Linux.
- **Bootable Pendrive:** Running a live Linux OS directly from a USB flash drive without altering local storage.
- **Virtual Machine (VM):** Running Linux as a virtualized guest OS inside a host operating system.

1. Virtual Machine Manager (Hypervisor)

- **Type-1 (Baremetal):** Runs directly on computer hardware without a host OS.
 - *Examples:* VMware ESXi, Hyper-V Server.
- **Type-2 (Hosted):** Runs as a software application inside an existing host operating system.
 - *Examples:* Oracle VirtualBox, VMware Workstation.

B) Working with Linux Commands

Command Explanations:

- **Terminal Basics:**
 - `pwd`: Prints the current working directory path.
 - `mkdir`: Creates a new directory.
 - `rmdir`: Deletes an empty directory.
 - `dir`: Lists directory contents.
 - `ls`: Lists files and folders.

- cd: Changes current directory.
- cd ..: Moves up to the parent directory.
- clear: Clears the terminal screen.
- history: Shows previously executed commands.
- man: Displays manual pages for a command.
- whoami: Displays active user name.

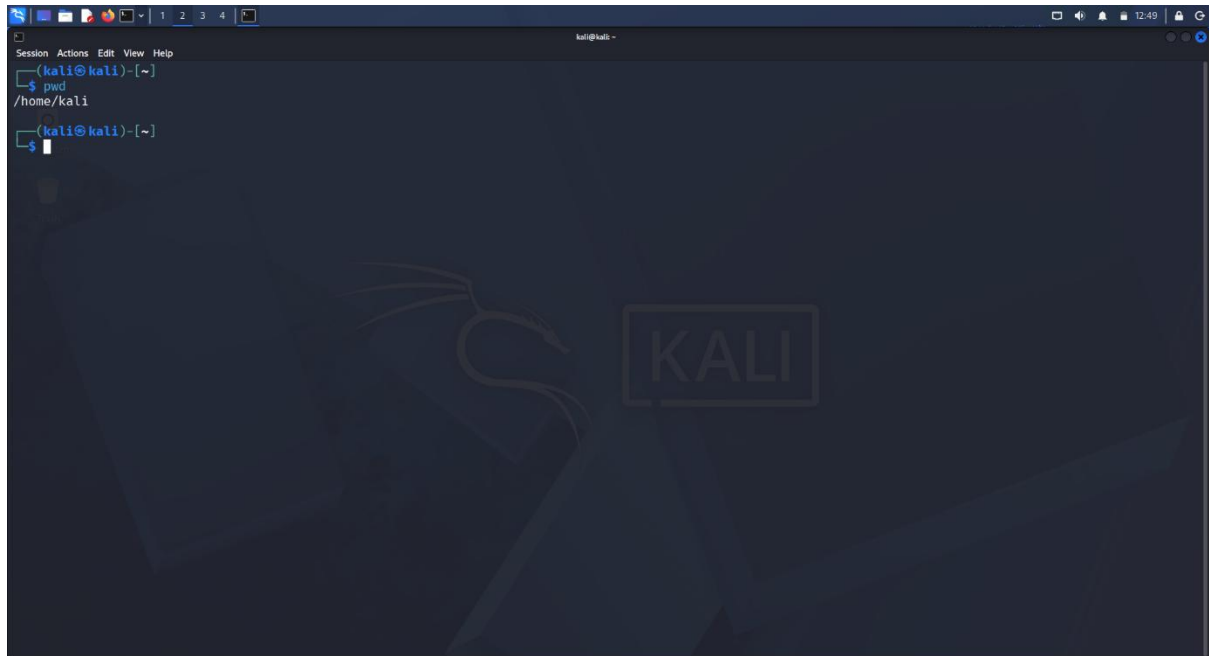
- **Files & Viewing:**

- vi: Terminal text editor.
- file: Identifies file type.
- rm: Removes files.
- cp: Copies files.
- mv: Moves/renames files.
- cat: Displays file contents.
- more: Scrolls through file contents downward.
- less: Navigates file contents bidirectionally.
- head: Displays initial lines of a file.
- tail: Displays final lines of a file.
- chmod: Alters file permissions.

- **Redirection:**

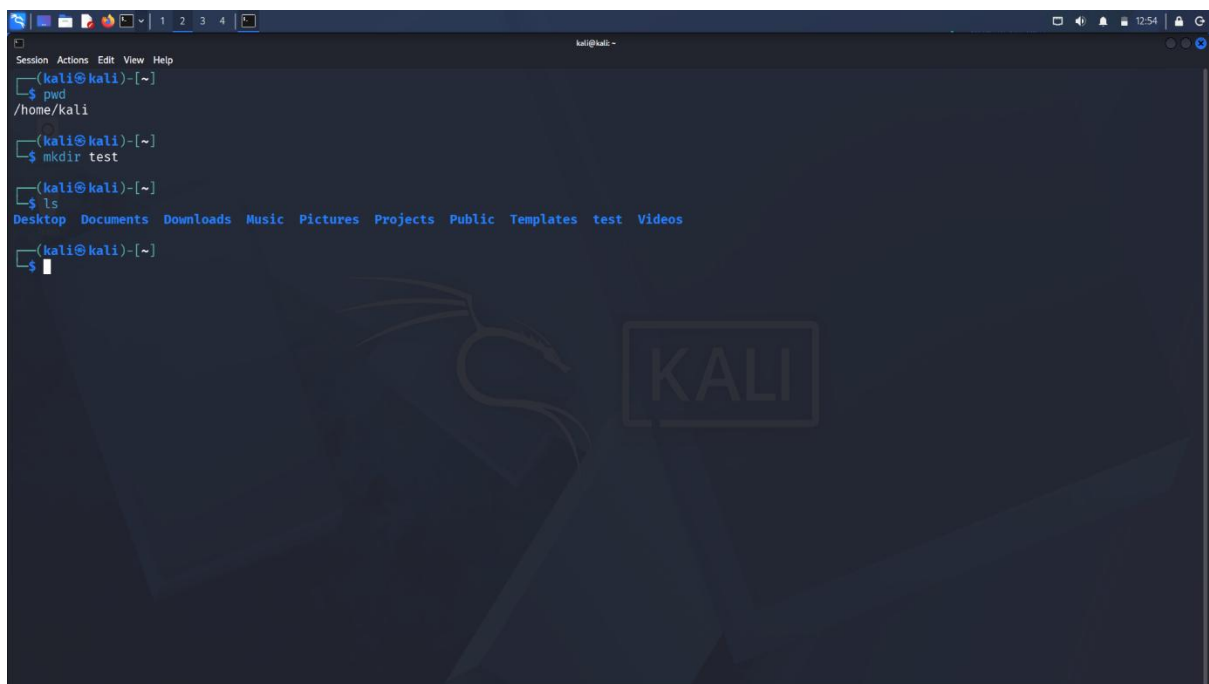
- >: Redirects output and overwrites file.
- >>: Redirects output and appends to file.

- <: Redirects file content as standard input.



A terminal window titled 'kali@kali' with a menu bar (Session, Actions, Edit, View, Help) and a tab bar (1, 2, 3, 4). The prompt is '(kali@kali)-[~]'. The user enters 'pwd' and the output is '/home/kali'. The prompt is then '(kali@kali)-[~]' with a cursor.

```
(kali@kali)-[~]  
$ pwd  
/home/kali  
(kali@kali)-[~]  
$
```



A terminal window titled 'kali@kali' with a menu bar (Session, Actions, Edit, View, Help) and a tab bar (1, 2, 3, 4). The prompt is '(kali@kali)-[~]'. The user enters 'mkdir test' and the output is '/home/kali'. The prompt is then '(kali@kali)-[~]'. The user enters 'ls' and the output is 'Desktop Documents Downloads Music Pictures Projects Public Templates test Videos'. The prompt is then '(kali@kali)-[~]' with a cursor.

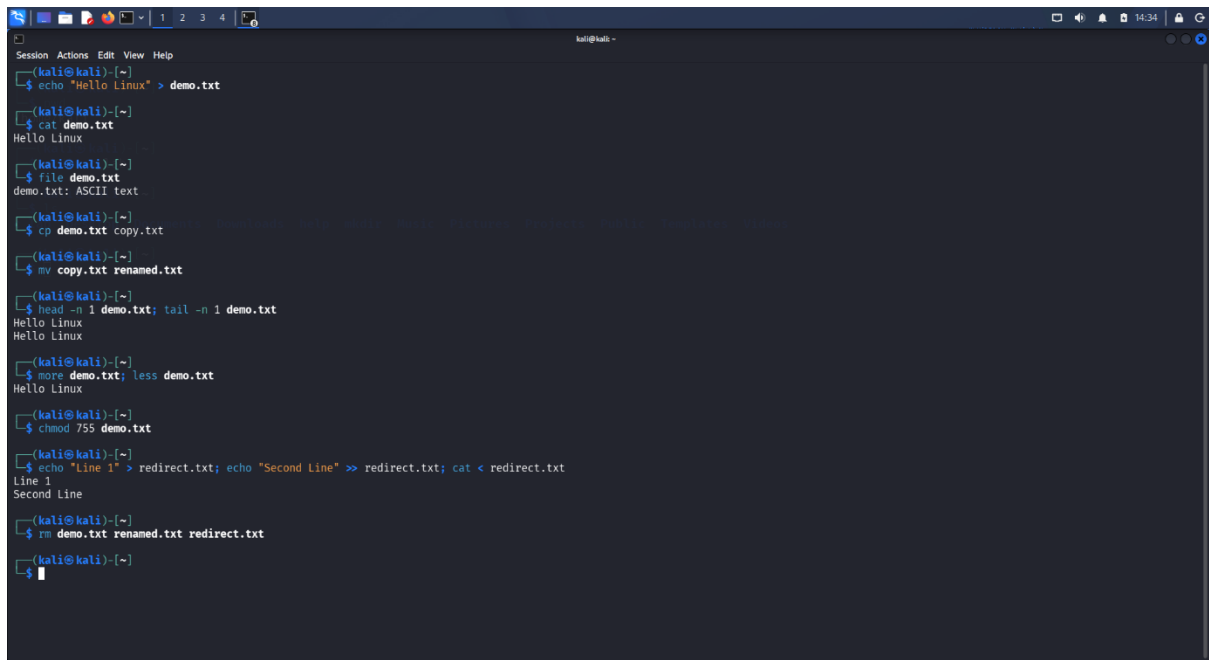
```
(kali@kali)-[~]  
$ pwd  
/home/kali  
(kali@kali)-[~]  
$ mkdir test  
(kali@kali)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Projects Public Templates test Videos  
(kali@kali)-[~]  
$
```

```
kali@kali: ~/c_lab
Session Actions Edit View Help
(kali@kali)~/c_lab
$ cd c_lab
cd: no such file or directory: c_lab
(kali@kali)~/c_lab
$
```

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)~/c_lab
$ cd c_lab
cd: no such file or directory: c_lab
(kali@kali)~/c_lab
$ cd ..
(kali@kali)~
$
```

```
Session  Terminal Emulator  Use the command line
(kali@kali)-[~/c_lab]
$ cd c_lab
cd: no such file or directory: c_lab
(kali@kali)-[~/c_lab]
$ cd ..
(kali@kali)-[~]
$ rmdir c_lab
(kali@kali)-[~]
$ dir
Desktop  Documents  Downloads  help  mkdir  Music  Pictures  Projects  Public  Templates  Videos  /usr/share/doc/ncftools/  /usr/share/doc/ncftools/ncftools-1.0.0-1
(kali@kali)-[~]
$
```

```
Session  Actions  Edit  View  Help
(kali@kali)-[~/c_lab]
$ cd c_lab
cd: no such file or directory: c_lab
(kali@kali)-[~/c_lab]
$ cd ..
(kali@kali)-[~]
$ rmdir c_lab
(kali@kali)-[~]
$ dir
Desktop  Documents  Downloads  help  mkdir  Music  Pictures  Projects  Public  Templates  Videos  /usr/share/doc/ncftools/  /usr/share/doc/ncftools/ncftools-1.0.0-1
(kali@kali)-[~]
$ whoami
kali
(kali@kali)-[~]
$ history
1  pwd
2  me/kali
3  (kali@kali)-[~]
4  $
5  (kali@kali)-[~]
6  pwd
7  mkdir test
8  ls
9  rmdir test
10 ls
11 pwd
12 cd c_lab
13 cd ..
14 rmdir c_lab
15 dir
16 whoami
(kali@kali)-[~]
$
```



```
Session Actions Edit View Help
kali@kali -
(kali@kali)~$ echo "Hello Linux" > demo.txt
(kali@kali)~$ cat demo.txt
Hello Linux
(kali@kali)~$ file demo.txt
demo.txt: ASCII text
(kali@kali)~$ cp demo.txt copy.txt
(kali@kali)~$ mv copy.txt renamed.txt
(kali@kali)~$ head -n 1 demo.txt; tail -n 1 demo.txt
Hello Linux
Hello Linux
(kali@kali)~$ more demo.txt; less demo.txt
Hello Linux
(kali@kali)~$ chmod 755 demo.txt
(kali@kali)~$ echo "Line 1" > redirect.txt; echo "Second Line" >> redirect.txt; cat < redirect.txt
Line 1
Second Line
(kali@kali)~$ rm demo.txt renamed.txt redirect.txt
(kali@kali)~$
```

Conclusion:

The Linux programming environment was successfully set up using Oracle VirtualBox, and essential Linux commands for terminal navigation, file manipulation, and output redirection were executed and verified.